UTM
UNIVERSITI TEKNOLOGI MALAYSIA

COURSE INFORMATION

School/Faculty:	**Faculty of Computing**		
Program name:	**Bachelor in Software Engineering with Honors**		
Course code:	**SCSE2243**	**Academic Session/Semester:**	**2025/2026-2**
Course name:	**Application Development Project 1**	**Pre/co requisite (course name and code, if applicable):**	**SCSE2123 Software Requirements Engineering SCSE 1224 Advanced Programming**
Credit hours:	**3**		

Course synopsis	Application Development Project I is a cornerstone design experience course that integrates software engineering knowledge and skills acquired from previous courses. Students work in teams to engage with the community to identify real-world problems, conduct observation and analysis, and propose feasible software-based solutions. The course combines teamwork and individual responsibilities, where requirements elicitation, analysis, and design are performed collaboratively, while individual contributions are clearly defined and documented in each deliverable. The project culminates in the development of a high-fidelity prototype and is conducted under realistic constraints of time and resources with firm deadlines, requiring teams to plan, set goals, and manage their work effectively. Overall, the course aims to expose students to standard tools and current techniques or technologies in software development, while developing their capabilities as effective team members and leaders in software development projects.
Course coordinator (if applicable)	Dr Mohd Yazid bin Idris

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Dr Muhammad Khatibsyarbini			
	Dr Alif Ridzuan			

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO*	PLO (MQF Cluster Code)	Taxonomies and Graduate attributes	T&L methods	Assessment methods
CLO1	Evaluate the needs of community for the innovative solutions	PLO2 (C2)	C6 (SC2)	POPBL Cornerstone Project Seminar	Community Engagement (5%), Proposal (5%), Project Progress (10%)

CLO2	Work in development team to develop high fidelity prototype with the involvement of diverse stakeholders in community	PLO4 (C3B)	P5 A(SI1)	Cornerstone Project	Project Management (5%), Project Progress (20%), High-Fidelity Prototype (20%), Peer Asessessment (5%), Final Report (10%)
CLO3	Discover potential entrepreneurial opportunities.	PL10 (C4B)	A3 A(ES5)	Product Demo, Presentation	Presentation (5%), Proposal (5%), Project Management (5%), High-Fidelity Prototype (5%)

<MQF Cluster Code>
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numercy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism
<Taxonomies>
Cognitive (C) - C1-C6, **Psychomotor (P)** - P1-P7, **Affective (A)** - A1-A5
<Graduate Attributes>
SC1, SC2, SC3 = Scholars, **SI1, SI2, SI3, SI4** = Social Intelligence, **IT1, IT2, IT3** = Innovative Talent, **CC1, CC2, CC3, CC4** = Constructive Citizen, **ES1, ES2, ES3, ES4, ES5** = Enterprising Skills,

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active learning	Conducted through in-class activities based on informal cooperative learning.
2	Project-oriented Problem-based (Po-Pb)	Conducted through a project from communities' requirement, which problems are solved throughout the semester. Students in a group of 3-4 are required to deliver a Software Engineering document and demonstrate high fidelity working prototype.
3		

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :
Leadership skills, Communication skills

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO* Compulsory	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			

Week 1	Application Development Foundation	CLO1					2			1	2	2	7
Week 2	Theme Selection & Ideation Community Engagement Project Proposal	CLO1					2			1	2	2	7
Week 3		CLO1					2			1	2	2	7
Week 4		CLO1 CLO2	2			1					2	2	7
Week 5	Project Management Tool (JIRA)	CLO2	2			1					2	2	7
Week 6	Management & Requirement Doc - Progress 1	CLO2	1		1	1					2	2	7
Week 7	Visual Modeling & App Design (Visual Paradigm)	CLO2	2			1					2	2	7
Week 8													
Week 9	Requirement & App Design Doc - Progress 2	CLO2	1		1	1					2	2	7
Week 10	Collaborative Front-End UI Design	CLO2	2			1					2	2	7
Week 11		CLO2			2	1					2	2	7
Week 12	High Fidelity Prototype - Progress 3	CLO2	1		1	1					2	2	7
Week 13		CLO2	1		1	1					2	2	7
Week 14	Final Report	CLO2 CLO3	2			1					2	2	7
Week 15	Demo Prototype	CLO3	2			1					2	2	7
											SUB-TOTAL SLT :		**98**

Continous Assessment		**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Community Engagement	5		2			2
2	Proposal / Planning	10	2			1	3
3	Progress 1	10	1			1	2
4	Progress 2	10	1			1	2
5	Progress 3	10	1			1	2
6	Peer Assessment	5			2		2

7	High Fidelity Prototype	25	2			1	3
8	Project Management	10	2			1	
9	Final Report	10	2			1	3
10	Demo & Presentation	5	2			1	3
					SUB-TOTAL SLT :		**22**

Summative Assessment		**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1							0
2							0
					SUB-TOTAL SLT :		**0**
					SLT for Assessment:		**22**
					GRAND TOTAL SLT:		**120**

A	% SLT for F2F Physical Component	37.40
B	% SLT for Online & Independent Learning Component :	62.60
C	%SLT for Online Component:	34.17
D	% SLT for All Practical Component:	4.88
D1	% SLT for F2F Physical Practical Component:	4.88
D2	% SLT for F2F Online Practical Component:	0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)
Software: Visual Paradigm, Cloud Application: Github, Modern Requirement (with Gen.AI), Jira

References (include required and further readings, and should be the most current)

Other additional information (if applicable)
Academic honesty and plagiarism: *(Below is just a sample)*

All tasks and are group activities (UNLESS EXPLICITLY INDICATED AS SINGLE PERSON ACTIVITIES) Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

Disclaimer:

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Prepared by: **Name:** **Signature:** **Date:**	**Certified by:** **Name:** **Signature:** **Date:**